INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI

IDC216 : Statistics for Scientists (2025-26) **Task 02 (Week 03 : Aug 18-22)**

---

1. **Boiling point of water at different pressures.** In an experiment you measure the boiling point of water at different atmospheric pressures and obtain the following readings:

Pressure (atm)	Boiling Point (°C)
0.5	81.3
0.8	93.5
1.0	100.0
1.2	104.8

Estimate the boiling point of water at 0.9 atm. Here 1 atm represents one unit of atmospheric pressure.

2. **Population growth of bacteria.** In your experiment with bacterial culture the following readings are obtained at different times:

Time (hours)	Number of Bacteria (in millions)
11:00 AM	2.0
12:00 noon	2.7
1:00 PM	3.6
2:00 PM	4.9

Estimate the population of bacteria at 1:30 PM. What would be the population at 2:30 PM?

3. **Cooling of a heated iron rod.** A heated iron rod is cooling in air. It temperature is measured at different times:

Time (minutes)	Temperature (°C)
0	200
5	163
10	137
15	Forgot to take reading
20	107

Estimate its temperature at $t = 15$ minutes. Based on this, now estimate the temperature at $t = 12$ minutes.

4. **Growth of a plant.** The height of a plant was recorded at different days:

Day	Height (cm)
0	4.0
2	5.5
4	7.2
6	9.4
8	12.1

Estimate the plant's height at day 5.

5. **Radioactive decay.** The activity of a radioactive sample (in counts per minute) was measured at different times:

Time (minutes)	Activity (cpm)
0	1280
5	900
10	640
15	450

Estimate the activity at $t = 8$ minutes. Guess the formula for estimating this activity at an arbitrary t. If you wish to compare it with the well-known radioactive decay formula $N(t) = N_0 e^{-\lambda t}$ then what value of λ would you like to assign to your radioactive sample.